Probability & Probability Distribution

Formative Assessment 8

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Section: APM110 - SECTION 2

Github Link: https://github.com/arvincunanan152-stack/APM1110-FA8.git

Problem 1

An urn has 10 red, 8 black, and 7 green balls, so there are 25 balls in total. One color is chosen at random, so each color has probability $\frac{1}{3}$. Then 4 balls are drawn. X is the number of the 4 balls that match the chosen color.

(a) Find $P(X = 0)$

I used the law of total probability and conditioned on the chosen color. The total number of ways to pick 4 balls is

$$\binom{25}{4} = 12650.$$

$X = 0$ means all 4 balls come from the balls that are not the chosen color.

- Red chosen: 15 non-red balls, so $P(X = 0 \mid \text{red}) = \frac{\binom{15}{4}}{\binom{25}{4}} = \frac{1365}{12650}$
- Black chosen: 17 non-black balls, so $P(X = 0 \mid \text{black}) = \frac{\binom{17}{4}}{\binom{25}{4}} = \frac{2380}{12650}$
- Green chosen: 18 non-green balls, so $P(X = 0 \mid \text{green}) = \frac{\binom{18}{4}}{\binom{25}{4}} = \frac{3060}{12650}$

Putting these together:

$$P(X = 0) = \frac{1}{3}\left(\frac{1365}{12650} + \frac{2380}{12650} + \frac{3060}{12650}\right) = \frac{1}{3} \cdot \frac{6805}{12650} = \frac{6805}{37950} \approx 0.1793$$

```
total <- choose(25, 4)
p_x0 <- (1/3) * (choose(15, 4) + choose(17, 4) + choose(18, 4)) / total
round(p_x0, 4)
```

```
## [1] 0.1793
```

Answer: $P(X = 0) \approx 0.1793$

(b) Find $P(X_i = 1)$ for $i = 1, 2, 3, 4$

Here $X_i = 1$ if the ith ball picked is the chosen color. Every ball in the urn is equally likely to be the ith pick, so once the color is known, $P(X_i = 1)$ is that color's share of the urn:

$$P(X_i = 1 \mid \text{red}) = \frac{10}{25}, \quad P(X_i = 1 \mid \text{black}) = \frac{8}{25}, \quad P(X_i = 1 \mid \text{green}) = \frac{7}{25}.$$

By the law of total probability:

$$P(X_i = 1) = \frac{1}{3} \cdot \frac{10}{25} + \frac{1}{3} \cdot \frac{8}{25} + \frac{1}{3} \cdot \frac{7}{25} = \frac{1}{3} \cdot \frac{25}{25} = \frac{1}{3}.$$

```
p_xi <- (1/3) * (10/25 + 8/25 + 7/25)
round(p_xi, 4)
```

```
## [1] 0.3333
```

Answer: $P(X_i = 1) = \frac{1}{3} \approx 0.3333$ for $i = 1, 2, 3, 4$

(c) Find $E[X]$

The count of matching balls is the sum of the indicators:

$$X = X_1 + X_2 + X_3 + X_4.$$

Each X_i is an indicator, so $E[X_i] = P(X_i = 1) = \frac{1}{3}$. By linearity of expectation:

$$E[X] = E[X_1] + E[X_2] + E[X_3] + E[X_4] = 4 \cdot \frac{1}{3} = \frac{4}{3}.$$

```
round(4 * p_xi, 4)
```

```
## [1] 1.3333
```

Answer: $E[X] = \frac{4}{3} \approx 1.3333$

Problem 2

The Yule-Simon distribution has

$$P(X = n) = \frac{4}{n(n+1)(n+2)}, \quad n \geq 1.$$

(a) Show that the probabilities sum to 1

First, every term is positive, so $P(X = n) \geq 0$. Next I used the hint:

$$\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right).$$

Check using $\frac{k}{n(n+k)} = \frac{1}{n} - \frac{1}{n+k}$ with $k = 2$ on the left side. Multiplying by 4 gives

$$P(X = n) = 2\left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right).$$

Let $a_n = \dfrac{1}{n(n+1)}$. Then $P(X = n) = 2(a_n - a_{n+1})$, so the partial sum telescopes:

$$\sum_{n=1}^{N} P(X = n) = 2(a_1 - a_{N+1}) = 2\left(\frac{1}{2} - \frac{1}{(N+1)(N+2)}\right).$$

Letting $N \to \infty$, the last term goes to 0:

$$\sum_{n=1}^{\infty} P(X = n) = 2 \cdot \frac{1}{2} = 1.$$

So this is a valid PMF. Numeric check:

```
n <- 1:1e6
pmf <- 4 / (n * (n + 1) * (n + 2))
round(sum(pmf), 4)
```

```
## [1] 1
```

(b) Show that $E[X] = 2$

$$E[X] = \sum_{n=1}^{\infty} n \cdot \frac{4}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)}.$$

Using $\frac{1}{(n+1)(n+2)} = \frac{1}{n+1} - \frac{1}{n+2}$, the partial sum telescopes:

$$\sum_{n=1}^{N}\left(\frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{1}{2} - \frac{1}{N+2} \to \frac{1}{2}.$$

Therefore

$$E[X] = 4 \cdot \frac{1}{2} = 2.$$

```
round(sum(n * pmf), 4)
```

```
## [1] 2
```

(c) Show that $E[X^2] = \infty$

$$E[X^2] = \sum_{n=1}^{\infty} n^2 \cdot \frac{4}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{n}{(n+1)(n+2)}.$$

For $n \geq 1$ we have $n+1 \leq 2n$ and $n+2 \leq 3n$, so $(n+1)(n+2) \leq 6n^2$. This gives

$$\frac{n}{(n+1)(n+2)} \geq \frac{n}{6n^2} = \frac{1}{6n}.$$

So

$$E[X^2] \geq \frac{4}{6}\sum_{n=1}^{\infty} \frac{1}{n}.$$

The harmonic series $\sum \frac{1}{n}$ diverges, so by the comparison test $E[X^2] = \infty$.

In R, the partial sums just keep growing:

```
partial <- cumsum(n^2 * pmf)
data.frame(
  terms = c(10, 1000, 1e5, 1e6),
  partial_sum = round(partial[c(10, 1000, 1e5, 1e6)], 4)
)
```

```
##   terms partial_sum
## 1 1e+01      4.7462
## 2 1e+03     21.9539
## 3 1e+05     40.3607
## 4 1e+06     49.5709
```

Problem 3

The density is $f(x) = a + bx^2$ for $0 \leq x \leq 1$ (and 0 otherwise), and $E[X] = \frac{3}{5}$.

Equation 1 (total probability is 1):

$$\int_0^1 (a + bx^2)\,dx = \left[ax + \frac{bx^3}{3}\right]_0^1 = a + \frac{b}{3} = 1.$$

Equation 2 (the mean):

$$E[X] = \int_0^1 x(a + bx^2)\,dx = \left[\frac{ax^2}{2} + \frac{bx^4}{4}\right]_0^1 = \frac{a}{2} + \frac{b}{4} = \frac{3}{5}.$$

Solving: From Equation 1, $a = 1 - \frac{b}{3}$. Substituting into Equation 2:

$$\frac{1}{2}\left(1 - \frac{b}{3}\right) + \frac{b}{4} = \frac{3}{5}$$

$$\frac{1}{2} - \frac{b}{6} + \frac{b}{4} = \frac{3}{5}$$

$$\frac{b}{12} = \frac{3}{5} - \frac{1}{2} = \frac{1}{10}$$

$$b = \frac{12}{10} = \frac{6}{5}.$$

Then

$$a = 1 - \frac{1}{3} \cdot \frac{6}{5} = 1 - \frac{2}{5} = \frac{3}{5}.$$

I solved the same system in R to check:

```
A <- matrix(c(1, 1/3,
              1/2, 1/4), nrow = 2, byrow = TRUE)
rhs <- c(1, 3/5)
sol <- solve(A, rhs)
names(sol) <- c("a", "b")
round(sol, 4)
```

```
##   a   b
## 0.6 1.2
```

```
f <- function(x) sol["a"] + sol["b"] * x^2
round(integrate(f, 0, 1)$value, 4)                    # should be 1
```

```
## [1] 1
```

```
round(integrate(function(x) x * f(x), 0, 1)$value, 4) # should be 0.6
```

```
## [1] 0.6
```

Answer: $a = \frac{3}{5} = 0.6000$ and $b = \frac{6}{5} = 1.2000$

Problem 4

X has mean μ and variance σ^2. Let

$$Y = \frac{X - \mu}{\sigma} = \frac{1}{\sigma} X - \frac{\mu}{\sigma}.$$

Expected value. Using $E[aX + c] = aE[X] + c$:

$$E[Y] = \frac{1}{\sigma} E[X] - \frac{\mu}{\sigma} = \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0.$$

Variance. Using $\mathrm{Var}(aX + c) = a^2 \mathrm{Var}(X)$ (adding a constant does not change variance):

$$\text{Var}(Y) = \left(\frac{1}{\sigma}\right)^2 \text{Var}(X) = \frac{\sigma^2}{\sigma^2} = 1.$$

As a check, I tried $\mu = 10$ and $\sigma = 3$ with a large random sample:

```
set.seed(1110)
x <- rnorm(100000, mean = 10, sd = 3)
y <- (x - 10) / 3
round(mean(y), 4)  # close to 0
```

```
## [1] 0.0061
```

```
round(var(y), 4)   # close to 1
```

```
## [1] 0.9967
```

Answer: $E[Y] = 0$ and $\text{Var}(Y) = 1$

Problem 5

$X \sim \text{Poisson}(2)$ and $Y \sim \text{Poisson}(3)$ are independent, and $W = X + Y$.

(a) MGF of W

For $N \sim \text{Poisson}(\lambda)$, the MGF is

$$M_N(t) = E[e^{tN}] = \sum_{k=0}^{\infty} e^{tk} \frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)}.$$

Since X and Y are independent, $E[e^{t(X+Y)}] = E[e^{tX}]\,E[e^{tY}]$, so

$$M_W(t) = M_X(t)\,M_Y(t) = e^{2(e^t-1)} \cdot e^{3(e^t-1)} = e^{5(e^t-1)}.$$

(b) Distribution of W

$M_W(t) = e^{5(e^t-1)}$ has the same form as the Poisson MGF with $\lambda = 5$. By the uniqueness property of MGFs (an MGF that exists near $t = 0$ determines the distribution),

$$W \sim \text{Poisson}(5).$$

(c) $E[W]$ and $E[W^2]$

I differentiate $M_W(t)$ and evaluate at $t = 0$.

First derivative, using the chain rule:

$$M'_W(t) = 5e^t \, e^{5(e^t-1)} = 5e^t M_W(t).$$

At $t = 0$: $M_W(0) = e^0 = 1$, so

$$E[W] = M'_W(0) = 5 \cdot 1 \cdot 1 = 5.$$

Second derivative, using the product rule:

$$M''_W(t) = 5e^t M_W(t) + 5e^t M'_W(t) = 5e^t M_W(t) + 25e^{2t} M_W(t).$$

At $t = 0$:

$$E[W^2] = M''_W(0) = 5 + 25 = 30.$$

I also used R to differentiate the MGF symbolically:

```
M_W <- expression(exp(5 * (exp(t) - 1)))
d1 <- D(M_W, "t")
d2 <- D(d1, "t")

e_w  <- eval(d1, list(t = 0))
e_w2 <- eval(d2, list(t = 0))

round(e_w, 4)
```

```
## [1] 5
```

```
round(e_w2, 4)
```

```
## [1] 30
```

```
round(e_w2 - e_w^2, 4)  # variance
```

```
## [1] 5
```

As a final check, $\text{Var}(W) = E[W^2] - (E[W])^2 = 30 - 25 = 5$, which matches the Poisson variance $\lambda = 5$.

Answer: $M_W(t) = e^{5(e^t-1)}$, $W \sim \text{Poisson}(5)$, $E[W] = 5$, $E[W^2] = 30$